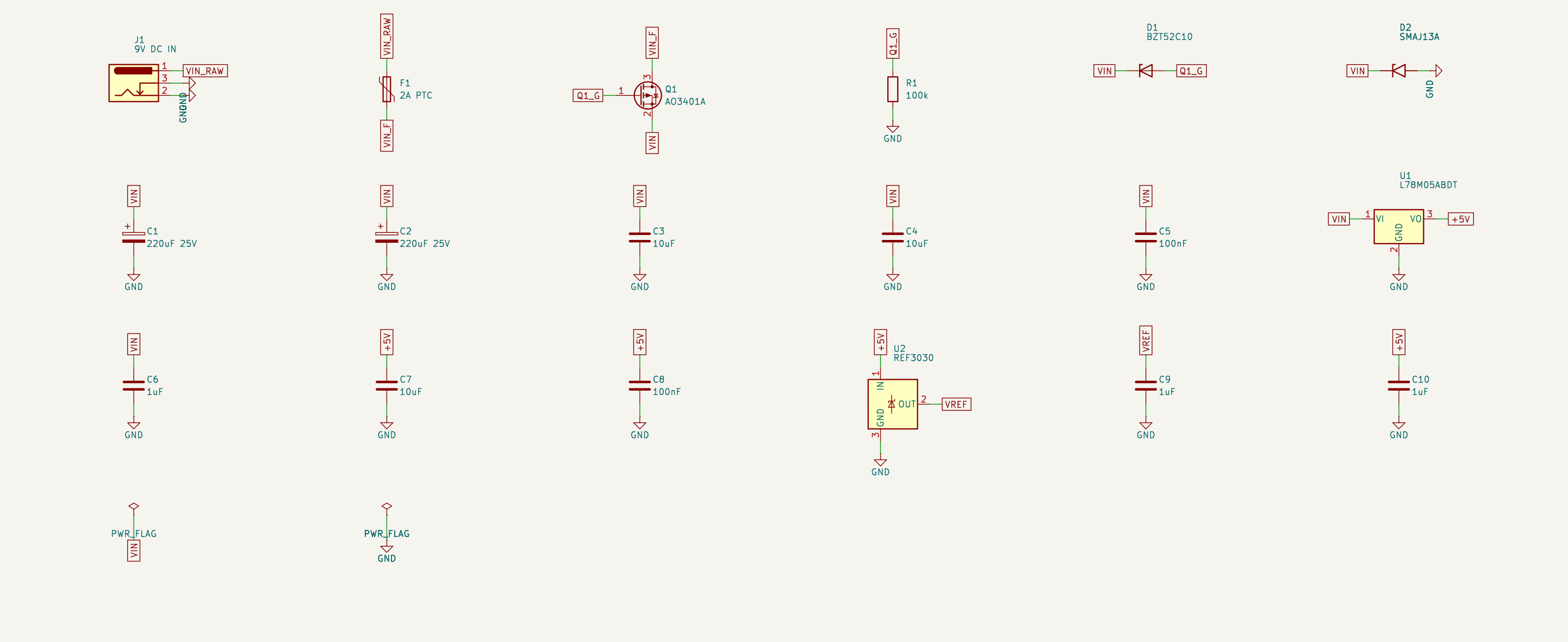
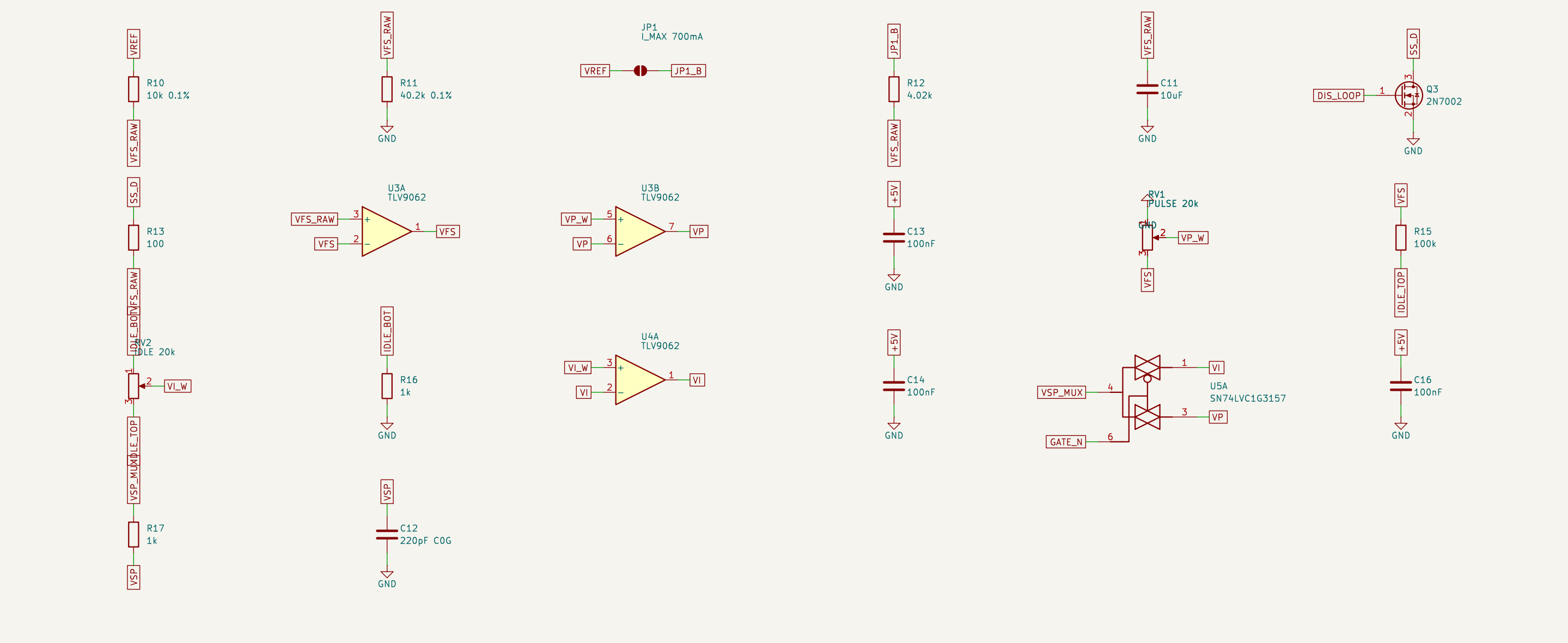
1 POWER INPUT, 5 V AND 3.000 V REFERENCE

9 V / >=1.5 A desktop adapter (5.5/2.1 mm, centre +).
Q1 = reverse-polarity protection, F1 = resettable fuse, D2 = TVS.



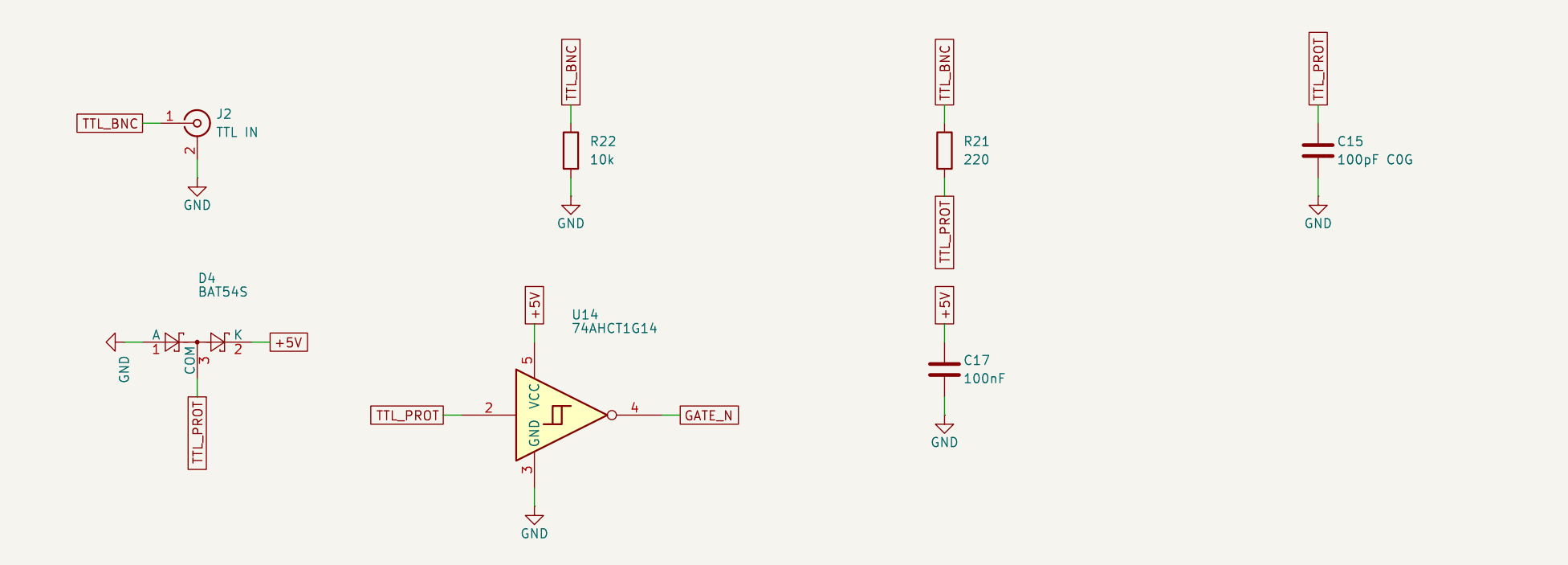
2 SETPOINT, SOFT START AND TTL GATING (scale 4 V/A)

VFS = 2.40 V = 600 mA full scale (JP1 closed: 2.80 V = 700 mA).
C11 + R10||R11 give an 80 ms soft start; Q3 discharges it when disabled.
U5 selects PULSE (TTL high) or IDLE (TTL low, 5..104 mA).



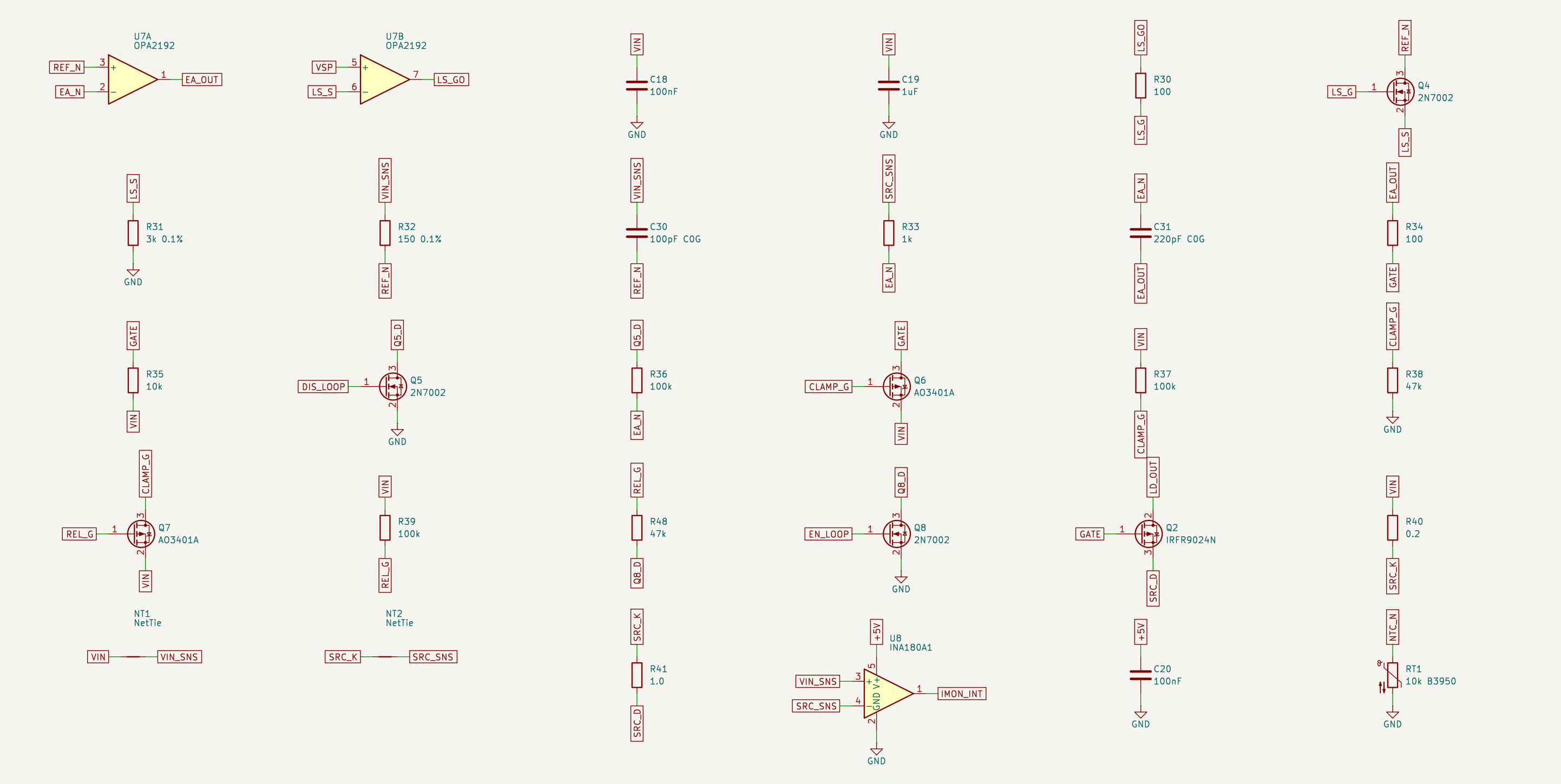
3 TTL INPUT (BNC, 10 k to GND, 3.3/5 V logic)

Schmitt trigger with TTL thresholds; unconnected input = IDLE.



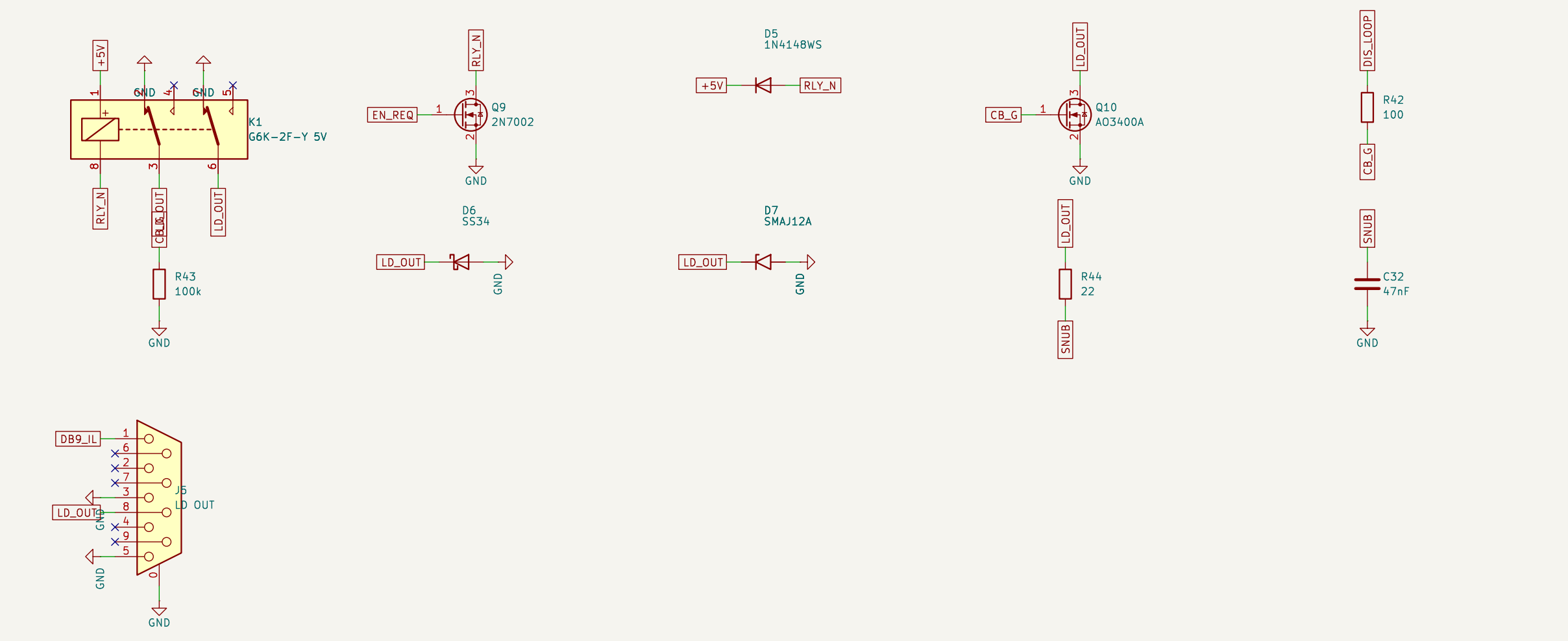
4 HIGH−SIDE CURRENT SOURCE (laser cathode at GND)

I_LD = VSP x (R32/R31) / R40 = VSP / 4 V/A.
U7A error amp senses R40 directly (Kelvin via NT1/NT2); R41 degenerates Q2.
Q6 clamps Q2 off unless EN_LOOP is high (fail−safe, Q7/Q8 release it).
Q5 forces U7A high while disabled so the integrator cannot wind up.



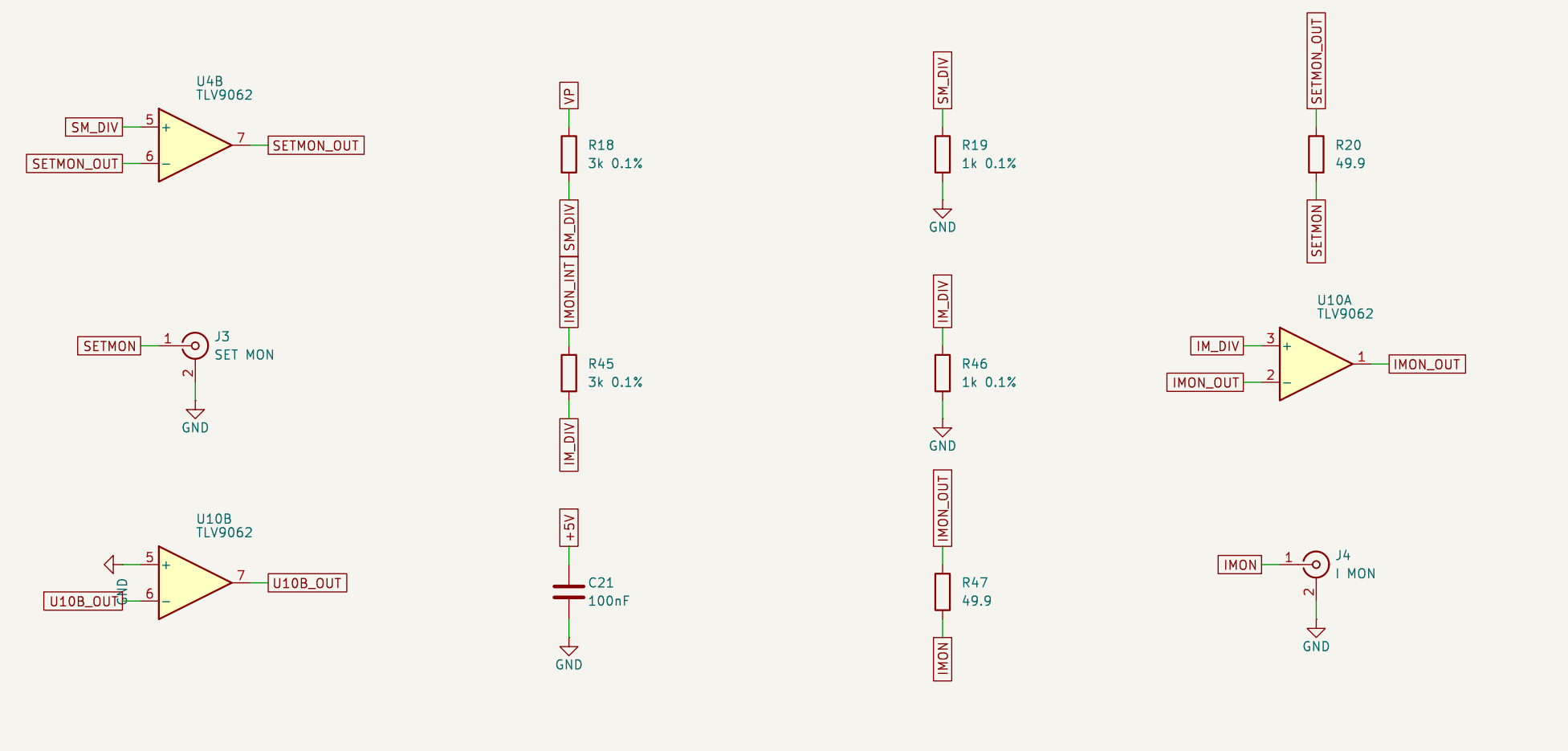
5 OUTPUT, SHORTING RELAY AND CROWBAR

K1 NC contacts short the laser whenever the driver is not armed or unpowered.
Q10 crowbar shorts it within 1 us of a fault; D6/D7 reverse and surge clamp.
DB9 follows Thorlabs LD−driver 'CG' pinout: 8 = anode, 3 = cathode/GND, 1−5 interlock.



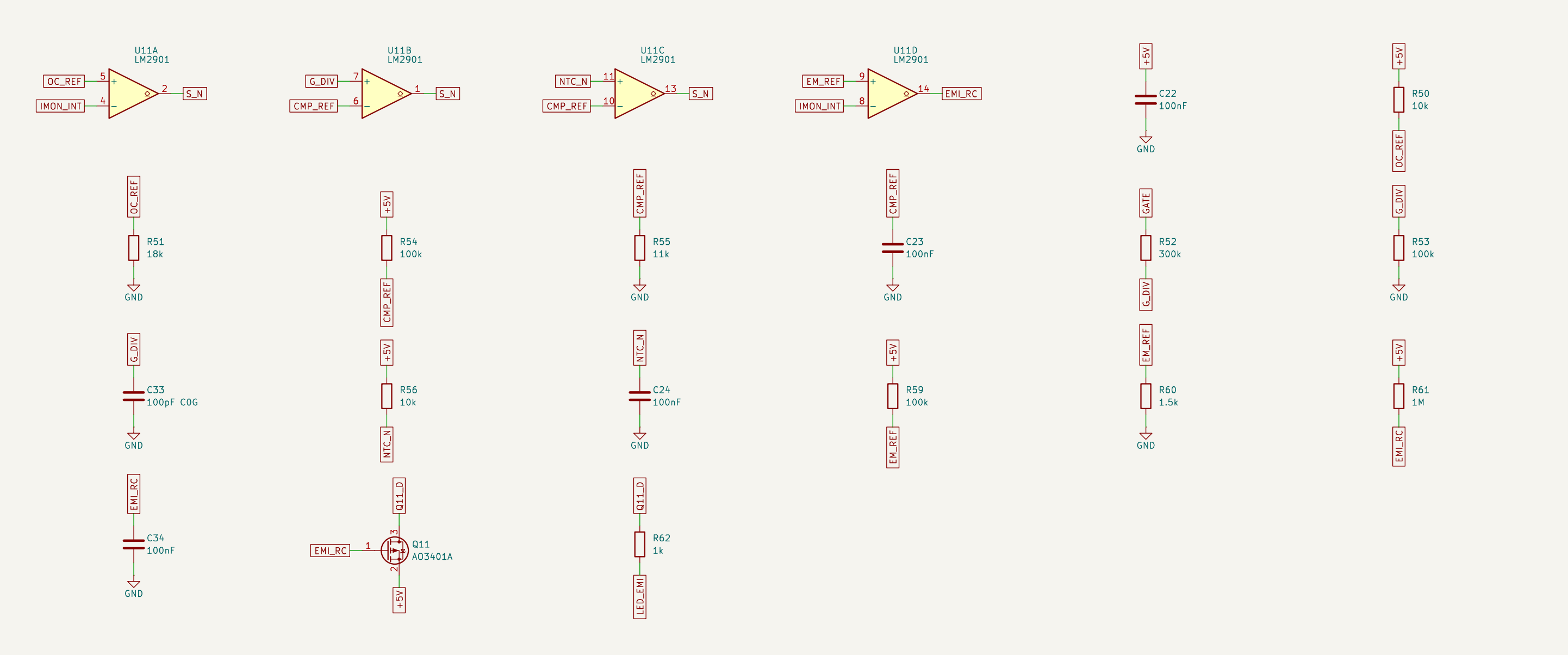
6 CURRENT AND SETPOINT MONITORS (1 V/A into high−Z)

I MON from INA180 (4 V/A) / 4; SET MON shows the PULSE setpoint / 4.



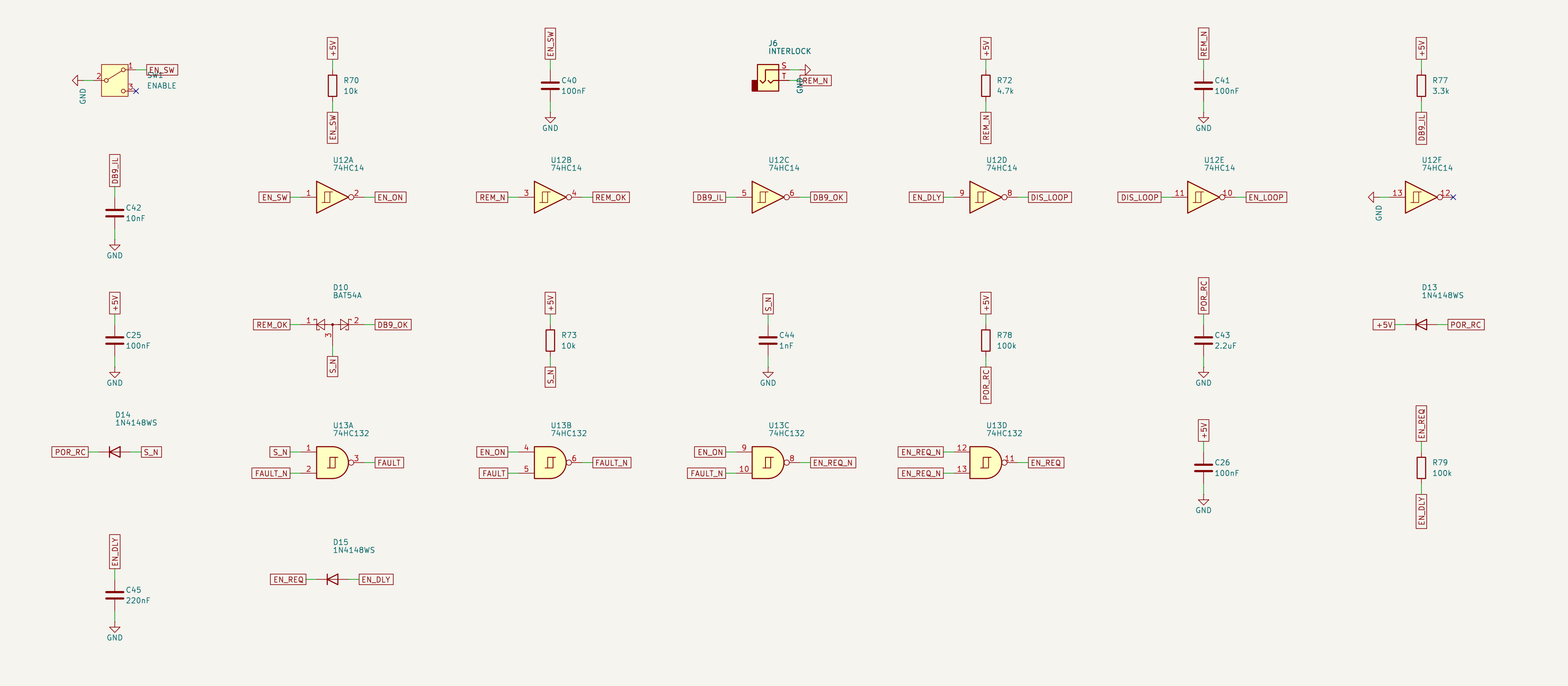
7 PROTECTION COMPARATORS

A: over−current > 800 mA. B: compliance / open laser (gate < 2 V for > 10 us).
C: Q2 heatsink > 85 C (NTC RT1). D: emission > 19 mA lights the EMISSION LED for ~0.15 s.

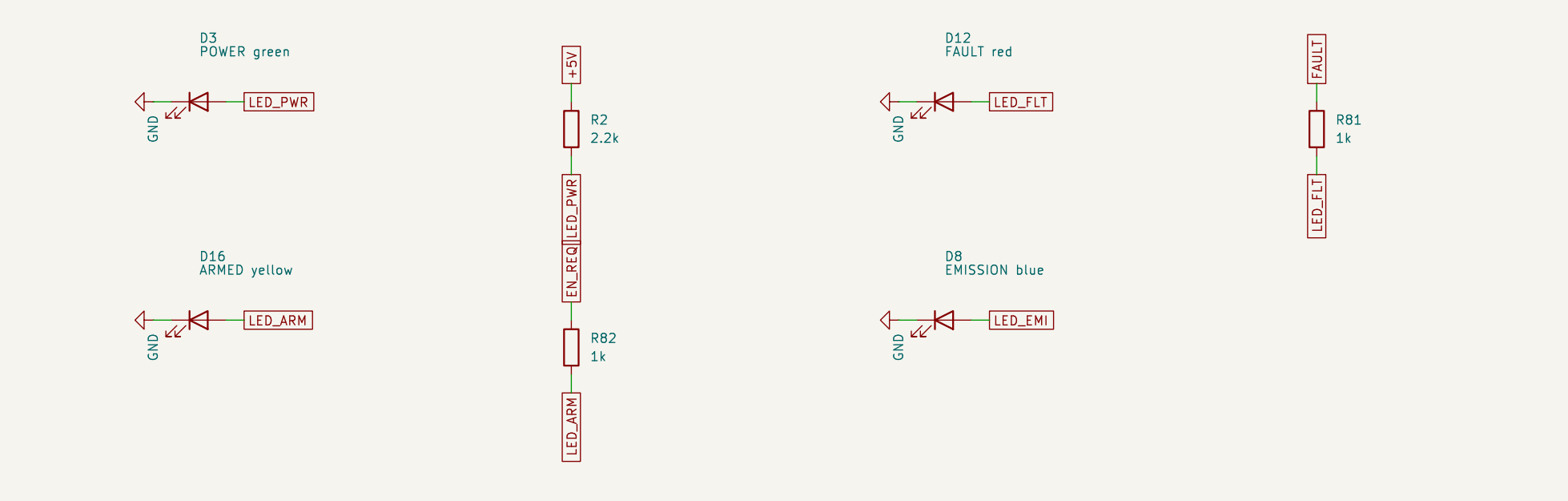


8 INTERLOCKS, FAULT LATCH AND ENABLE SEQUENCING

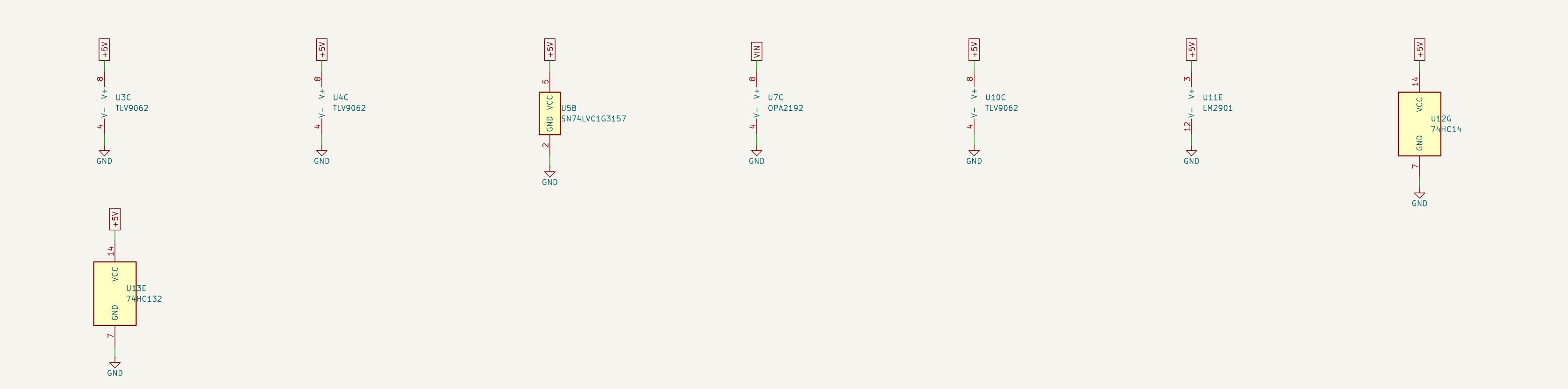
Any fault, open interlock or power−up sets the FAULT latch; reset by switching ENABLE off.
EN_REQ arms relay K1 at once; EN_LOOP releases the current source ~17 ms later.



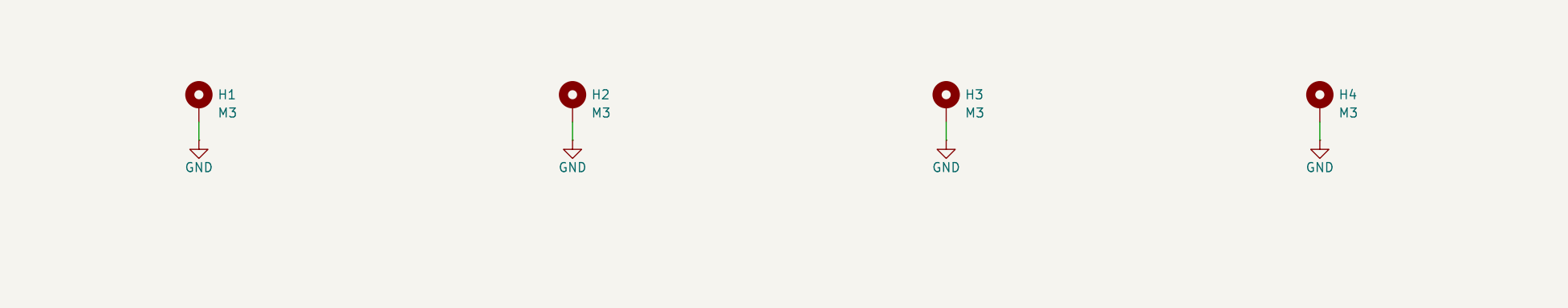
9 FRONT−PANEL INDICATORS



10 IC POWER PINS



11 MOUNTING HOLES



CLASS 4 LASER − read README safety section before use
Laser: DTR BDR−209 405 nm (500 mA nominal, 600 mA max), CATHODE AT GND
Generated from tools/design.py by tools/gen_sch.py
Ion−trap lab photoionisation 405 nm driver
Sheet: /
File: ld405−driver.kicad_sch
Title: 405 nm laser diode pulse driver (BDR−209)
Size: A0
Date: 2026−10−28
Rev: 1.0
KiCad E.D.A. 9.0.9
Id: 1/1